

ENONTEKIO NAKKALA, Finland

WMO: 027260

Lat: 68.6031N

Lon: 23.5761E

Elev: 374

StdP: 96.91

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.8	-28.0	-34.1	0.2	-30.7	-30.9	0.2	-27.9	11.3	-5.8	10.4	-6.4	2.2	0	0.792

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	23.2	15.4	20.9	14.3	18.6	13.4	16.6	21.1	15.4	19.3	14.2	17.4	3.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	11.1	17.6	13.7	10.2	16.5	12.7	9.5	15.8	47.9	21.4	44.3	19.6	41.1	17.5	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	8.3	7.3	-35.1	24.8	3.5	2.1	-37.6	26.4	-39.7	27.6	-41.6	28.8	-44.2	30.4		
			-35.1	17.8	3.3	1.9	-37.5	19.1	-39.4	20.2	-41.3	21.3	-43.7	22.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.3	-13.8	-13.2	-9.4	-3.2	3.9	9.2	13.1	10.8	6.5	-1.6	-8.0	-10.7
	DBStd	10.74	6.90	7.63	5.93	4.52	4.61	4.01	3.73	3.35	3.24	4.55	6.03	6.58
	HDD10.0	4342	739	651	601	397	197	63	12	31	111	359	539	643
	HDD18.3	7174	998	884	859	647	446	276	168	236	354	617	789	901
	CDD10.0	214	0	0	0	0	8	37	107	55	7	0	0	0
	CDD18.3	6	0	0	0	0	0	0	5	1	0	0	0	0
	CDH23.3	57	0	0	0	0	4	4	43	6	0	0	0	0
	CDH26.7	4	0	0	0	0	0	0	4	0	0	0	0	0
Wind	WSAvg	3.4	3.4	3.5	3.8	3.5	3.7	3.9	3.5	3.2	3.5	3.1	3.0	3.3
Precipitation	PrecAvg	469	25	22	16	20	36	57	76	64	49	43	33	27
	PrecMax	655	65	34	25	35	79	100	133	134	88	118	57	51
	PrecMin	353	5	8	7	6	8	18	25	27	15	13	9	6
	PrecStd	88	13	7	6	8	21	22	29	32	21	24	14	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.2	2.9	3.1	9.5	22.6	23.8	27.0	23.8	16.3	9.0	4.2	2.6
		MCWB	0.2	1.6	1.0	5.2	12.4	14.8	17.4	16.8	11.8	7.3	3.9	1.4
	2%	DB	-1.5	0.0	1.4	6.1	17.6	20.6	24.3	21.0	14.4	7.2	1.5	0.4
		MCWB	-1.9	-0.7	-0.1	2.7	10.3	13.0	16.1	14.7	11.1	6.2	1.1	0.0
	5%	DB	-3.3	-1.5	0.0	4.5	14.5	18.2	22.2	18.2	12.8	5.6	0.2	-0.8
		MCWB	-3.8	-2.0	-1.1	1.9	8.8	11.7	15.2	13.7	10.4	4.8	0.0	-1.2
	10%	DB	-5.0	-3.0	-1.2	3.0	11.3	15.7	19.7	16.3	11.4	4.1	-0.6	-1.8
		MCWB	-5.3	-3.5	-2.2	1.0	6.9	10.8	14.4	12.9	9.5	3.4	-0.8	-2.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.3	1.9	1.1	5.3	13.0	17.1	18.5	18.4	13.5	7.6	4.1	1.6
		MCDB	1.1	3.0	2.3	9.1	21.4	22.4	23.1	21.3	14.8	8.4	4.2	2.4
	2%	WB	-1.9	-0.6	0.1	3.2	11.2	14.7	17.2	15.8	11.9	6.4	1.1	0.0
		MCDB	-1.5	0.0	1.1	5.5	16.7	18.1	22.3	18.9	13.8	7.1	1.5	0.4
	5%	WB	-3.7	-1.9	-0.9	2.1	8.9	13.1	16.0	14.4	10.8	4.8	0.0	-1.0
		MCDB	-3.3	-1.5	-0.1	4.2	14.3	16.4	20.5	17.2	12.3	5.5	0.1	-0.8
	10%	WB	-5.3	-3.5	-2.1	1.2	7.2	11.5	15.1	13.3	9.9	3.5	-0.8	-2.1
		MCDB	-5.0	-3.0	-1.2	2.8	10.9	15.3	19.0	15.6	11.1	4.1	-0.7	-1.8
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.0	9.4	8.3	7.9	7.9	8.3	8.1	6.5	5.4	6.7	7.7
		MCDBR	7.7	7.4	7.0	8.5	13.2	11.8	12.2	11.0	8.2	5.5	4.6	5.2
		MCWBR	7.6	7.0	5.9	5.9	6.8	5.8	5.1	5.5	5.2	4.4	4.5	5.0
	5% WB	MCDBR	7.5	7.2	6.6	7.6	12.4	9.8	10.1	9.5	6.7	5.1	4.6	5.0
		MCWBR	7.4	6.8	5.7	5.3	6.7	5.2	4.8	5.2	4.6	4.2	4.5	4.8
Clear-Sky Solar Irradiance	taub	0.180	0.236	0.276	0.305	0.324	0.329	0.336	0.321	0.290	0.251	0.182	N/A	
	taud	1.916	2.086	2.188	2.217	2.391	2.493	2.486	2.508	2.560	2.286	1.966	N/A	
	Ebn at Noon	76	546	753	838	867	874	853	827	769	540	65	0	
	Edh at Noon	6	44	76	100	96	90	88	76	55	36	5	0	
All-Sky Solar Radiation	RadAvg	0.03	0.45	1.95	3.95	5.05	4.91	4.61	3.26	1.75	0.56	0.08	0.00	
	RadStd	0.00	0.04	0.08	0.18	0.48	0.35	0.45	0.23	0.17	0.08	0.01	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page